



Drivers of coastal benthic communities in a complex environmental setting

Yuting Vicky Lin^a, Pierre-Alexandre Château^b, Yoko Nozawa^{c,d,e}, Chih-Lin Wei^a,
Rainer Ferdinand Wunderlich^{f,g}, Vianney Denis^{a,*}

^a Institute of Oceanography, National Taiwan University, Taipei 10617, Taiwan

^b Department of Marine Environment and Engineering, National Sun Yat-Sen University, Kaohsiung 80420, Taiwan

^c Tropical Biosphere Research Center, University of the Ryukyus, Okinawa 905-0227, Japan

^d Biodiversity Research Center, Academia Sinica, Taipei 11529, Taiwan

^e Department of Marine Science, Faculty of Fisheries and Marine Science, Universitas Diponegoro, Semarang 50275, Indonesia

^f Department of Bioenvironmental Systems Engineering, National Taiwan University, Taipei 10617, Taiwan

^g INRAE, UR EABX, 33612 Cestas, France

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ABSTRACT

Analyzing the environmental factors affecting benthic communities in coastal areas is crucial for uncovering key factors that require conservation action. Here, we collected benthic and environmental (physical-chemical-historical and land-based) data for 433 transects in Taiwan. Using a *k*-means approach, five communities dominated by crustose coralline algae, turfs, stony corals, digitate, or bushy octocorals were first delineated. Conditional random forest models then identified physical, chemical, and land-based factors (e.g., light intensity, nitrite, and population density) relevant to community delineation and occurrence. Historical factors, including typhoons and temperature anomalies, had only little effect. The prevalent turf community correlated positively with chemical and land-based drivers, which suggests that anthropogenic impacts are causing a benthic homogenization. This mechanism may mask the effects of climate disturbances and regional differentiation of benthic assemblages. Consequently, management of nutrient enrichment and terrestrial runoff is urgently needed to improve community resilience in Taiwan amidst increasing challenges of climate change.

1. Introduction

Anthropogenic and natural stressors are causing changes in marine coastal benthic communities (hereafter communities) that alter the functions and services normally associated with pristine communities (Reverter et al., 2022). In coral reefs, the shift in dominance from corals to macroalgae typifies this degradation and probably represents the best-known trajectory of reefs in the Anthropocene (Hughes et al., 2017). This transition has served as a foundation for developing theoretical frameworks to explain reef dynamics following disturbances (Reverter et al., 2022). However, the dominance of macroalgae is not necessarily a sign of degradation (Cannon et al., 2023). The coral-macroalgal shift is geographically confined and shifts toward the dominance of taxa other than macroalgae are also common (Bell et al., 2022; Reverter et al., 2022; Tebbett et al., 2023). The reasons for the existence of different benthic trajectories remain unclear, while the original composition of the communities and their relationships with abiotic factors are probably the key to understanding this heterogeneity

(Lin and Denis, 2019).

The composition of reef communities is determined by a number of biophysical-chemical factors (Montaggioni and Braithwaite, 2009). Consequently, communities are usually organized to follow the spatial hierarchy of variables that independently or interactively control species distribution (Montaggioni and Braithwaite, 2009; Elahi et al., 2022). For example, Spalding et al. (2007) categorized the global ocean into twelve realms based on the shared evolutionary history of species, which generally matched broad-scale historical isolation events and thermal regimes. Light, sea surface temperatures (SST), and aragonite saturation tend to vary among biogeographic regions and are, therefore, crucial for the interpretation of large-scale benthic variability (Kleypas et al., 1999; Cresswell et al., 2017). At the regional scale (100–1000 km), chlorophyll *a*, nutrients, SST, turbidity, and wave exposure explain parts of the benthic variation (De'ath and Fabricius, 2008; Zinke et al., 2018; Jouffray et al., 2019). Environmental factors such as substrate availability, sedimentation, and wave motion are responsible for fine-scale benthic heterogeneity (1–100 km) (Sherman et al., 2016; Hsiao et al.,

* Corresponding author at: Institute of Oceanography, National Taiwan University, No. 1, Section 4, Roosevelt Road, Da'an District, Taipei 10617, Taiwan.
E-mail address: vianneydenis@ntu.edu.tw (V. Denis).

2021; Lange et al., 2021). In addition to the spatial scale, the different responses (e.g., thresholds) of communities to these drivers emphasize the importance of considering nonlinear relationships when studying their effects on community distribution and dynamics (Cutler et al., 2007; Gove et al., 2015). Meanwhile, human activities and natural disturbances can further complicate the spatial patterns created by these drivers and alter community composition (Robinson et al., 2018; Andrello et al., 2021).

Disturbances are of crucial importance in explaining the difference in reef community configurations, especially in the Anthropocene, where there is an increasing documentation of degraded communities (Montagioni and Braithwaite, 2009). Global temperature increases trigger recurrent mass coral bleaching and phase shifts in all reef areas (Hughes et al., 2018; Bell et al., 2022), and drive the poleward expansion of some tropical species in biogeographic transition zones (Zarzychny et al., 2023). At the regional scale, cyclones cause rapid changes in water temperature, turbidity, and wave velocity (Nadaoka et al., 2001), while coastal human activities and pollution alter nutrient content, sedimentation, chlorophyll *a*, and substrate types, leading to acute shifts in community composition (Fabricius et al., 2005; Sherman et al., 2016). In this context, it may be useful for spatial planning to link the spatial variation of communities to biophysical-chemical-historical-anthropogenic factors to identify the specific areas and drivers that deserve actions (Nolan et al., 2021). This is particularly important for countries that span biogeographical regions with different biological baselines, making the application of a single management approach unrealistic (Schiettekatte et al., 2022). Disentangling anthropogenic effects from the existing reef configuration in such areas also helps to predict future community trajectories, which could in turn benefit the long-term implementation of marine protected areas (Robinson et al., 2018). Nevertheless, few studies have explored the environmental factors underlying different reef systems across biogeographic regions that can simultaneously provide the leverage needed for effective management strategies (Reverter et al., 2022). Here, we used Taiwan – a country straddling various biogeographical regions and exposed to multiple stressors – as an example to examine the effects of environmental factors on the benthic communities. Specifically, we 1) identify the key drivers among 21 biophysical, chemical, historical, and land-based variables that underlie the community delineation; 2) examine the influential drivers that determine community delineation and 3) occurrence. Based on these results, we provide guidance for conservation in Taiwan by highlighting the effects of key drivers that could be locally addressed to improve community conditions and ultimately build more resilient reefs to climate change.

2. Materials and methods

2.1. Study locations

Taiwan's unique geographical location at the convergence of plate boundaries and ocean currents creates a particularly complex environment (Dai and Cheng, 2020). It reflects the island's location straddling three marine ecoregions (the East China Sea, the South China Sea, and South Kuroshio) (Spalding et al., 2007; Dai and Chen, 2018), resulting in a sharp decline in tropical marine species from south to north (e.g., Dai and Cheng, 2020; Dai, 2021). In this context, distinct communities emerge, influenced by regional and local environmental factors (Lin and Denis, 2019; Hsiao et al., 2021; Lin et al., 2022). Exacerbated by the Kuroshio Current, ocean warming and typhoons are increasingly impacting communities (Mei et al., 2015). The intensification of warming designates Taiwan as a hotspot of tropicalization, where the expansion of the tropical species' distribution could play a critical role in driving community changes (Lin et al., 2022). However, human activities related to the development and utilization of coastal areas have contributed markedly to the decline of coastal habitats for at least five decades (Yang et al., 1975; Dai, 1993; Meng et al., 2008). Although

Taiwan is an ideal place to study the impacts of both the oceanographic context and land-based activities on coastal communities, a comprehensive nation-wide assessment that could serve as a basis for marine conservation efforts is lacking.

A total of 87 locations in eight regions (Dongsha, East, Kenting, Lanyu, Ludao, North, Penghu, and Xiaoliuqiu) were selected for benthic surveys between 2016 and 2020 in Taiwan and its offshore islands (Fig. 1a). The sampling locations included the three ecoregions known in Taiwan, tropical and subtropical latitudes (~ 700 km in range), and shallow and mesophotic zones (5–40 m in depth). The present study focused on communities of sessile benthos that develop on hard substrates, therefore, omitted areas of sandy bottoms on the west coast of Taiwan.

2.2. Ecological surveys

At each location, five 20 m transect lines were laid at a constant depth with a minimum distance of 5 m between pairs of transect lines to capture the benthic heterogeneity that may exist at a fine spatial scale. These within-site differences have previously been assessed as relevant to local variations in substrate availability (Hsiao et al., 2021). The benthic composition of each transect was examined using 21 replicate photographs (0.5 × 0.5 m) taken at one-meter intervals along the transect line, and all images were automatically annotated using CoralNet 1.0 (Beijbom et al., 2015). A stratified sampling strategy was used to annotate benthic images by overlaying a 5 × 5 grid and identifying benthic organisms underneath a randomly positioned point in each cell. Labeling was performed using a deep neural network trained on 2750 photographs with 229 labels combining taxonomic (mostly genus) and morphological information on benthic organisms (Global ID: 22264; accuracy 63 %; Appendix A). The final decision on the label was made based on the relative probability of the top five identifications suggested by the algorithm, since this original procedure reduced 8.3 % of the errors when estimating covers compared to use only the most likely identification (Appendix B). The resulting 122 labels were later aggregated into morpho-functional categories (see Lin and Denis, 2019) to describe the benthic composition while substrate variables were removed to focus on the community only. In the final benthic matrix, the covers of 27 morpho-functional categories were recorded, and distributed across 433 transects (two transects are missing for technical reasons).

2.3. Environmental factors

A total of 21 physical, chemical, historical, and land-based factors were selected based on their potential impacts on the occurrence and abundance of benthic organisms (Table 1; references therein for justifications). The assessment of physical, chemical, and land-based factors focused on their chronic effects on communities, so calculations were based on a long-term period (preferably about 10 years, but depending on data availability) prior to sampling (Appendix C). For the historical factors, we examined the acute impacts of the most extreme typhoons and thermal anomaly events, thus providing information on their intensity, frequency, and time of recovery (Appendix C). Each transect was described in terms of eight physical, three chemical, five historical, and five land-based factors. A complete list of these possible drivers indicating resolution, time period, sources of data, and computation is available in Appendix C.

2.4. Community delineation

To find the optimal number of partitions (i.e., communities) among the 433 transects that maximize between-group heterogeneity while minimizing within-group heterogeneity, a *k*-means cascade was performed on the Hellinger-transformed benthic matrix. A decision on the optimal partitioning was made by the consensus of 26 clustering indices

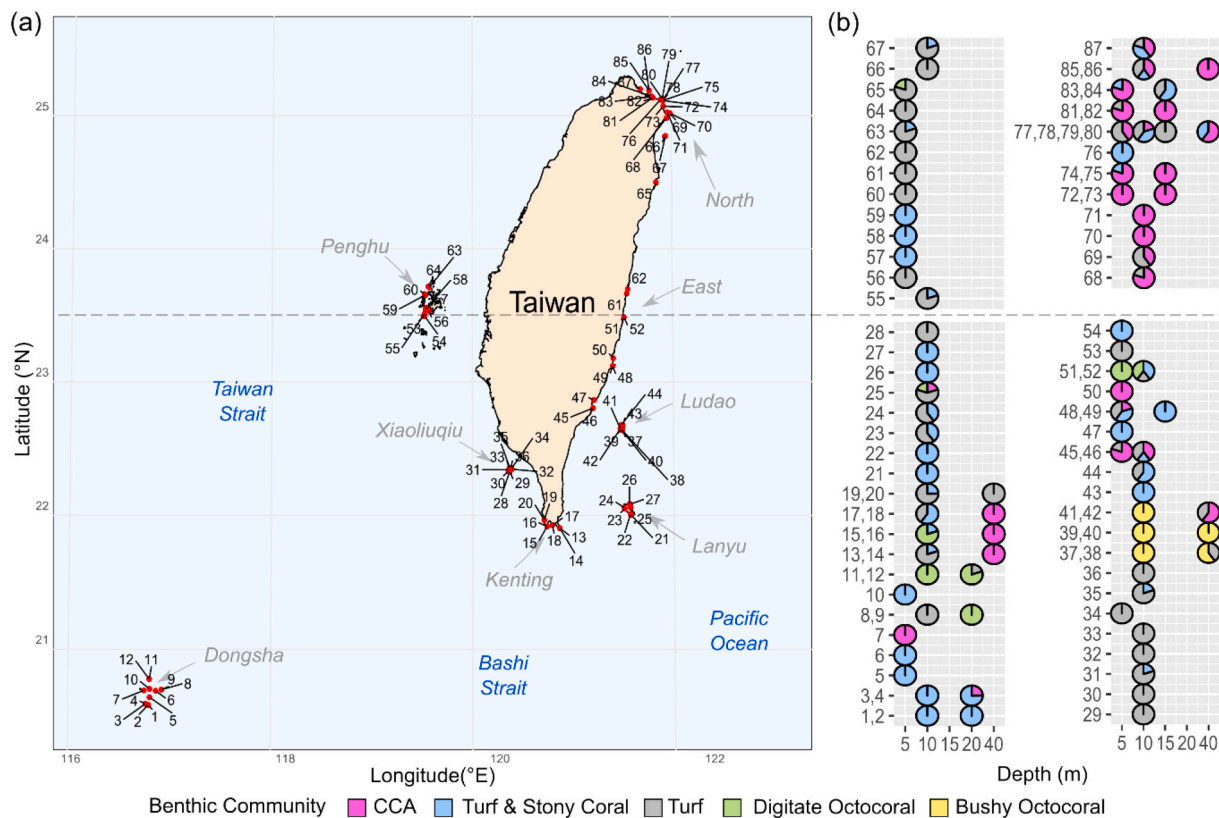


Fig. 1. 87 survey locations and their associated benthic communities.

(a) The 87 locations are distributed across eight regions (indicated in grey) around Taiwan. (b) The benthic communities to which each of the 433 transects belong. Numbers represent locations arranged according to increasing latitude. Horizontal dash lines indicate the Tropic of Cancer (23.5°N). Map projection: Mollweide. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

(Charrad et al., 2012). The resulting partitioning was then tested under 9999 permutations. The stability of transects assigned to a particular partition was assessed using bootstrapping and considered as the probability of occurrence for a particular partition (Yu et al., 2018). The Dufrene-Legendre indicator species analysis was used to select morpho-functional groups representative of each community according to their fidelity and specificity on transects (Dufrene and Legendre, 1997). A Moran's I test was used to test for spatial autocorrelation in the distribution of communities (Gittleman and Kot, 1990). Significant results can be interpreted as spatial patterns that are more dispersed (coefficient close to -1) or more clustered (coefficient close to +1) than a random distribution (Gittleman and Kot, 1990). Using a Hellinger-transformed Euclidean distance matrix, non-metric multidimensional scaling (nMDS) was used to visualize the relationship between transects assigned to different communities (Legendre and Gallagher, 2001), with benthic categories represented as transect scores weighted by their covers.

2.5. Environmental drivers of communities

For environmental factors with skewed distributions, a logarithm transformation was performed to reduce the impact of extreme values. Random Forest is a supervised learning algorithm that functions as an ensemble of numerous classification and regression trees, each fitted to a subset of the data generated by a bootstrap procedure (Breiman, 2001). It generally has high modeling performance and specializes in examining nonlinear relationships between the explained and response variables, which are prevalent in ecology (Cutler et al., 2007; Wunderlich et al., 2022). While random forests are known for an unbiased assessment of variable importance for correlated variables that may represent different ecological meanings, we adopt a conditional permutation

scheme (i.e., conditional random forest; CRF) to even better reflect the true impact of environmental factors (Cutler et al., 2007; Strobl et al., 2008). We constructed two types of CRF. A general CRF was constructed to examine which environmental factors were more important for the distribution of communities, with environmental factors as explanatory variables and partitioning of communities as response variables. An individual CRF was then constructed for each community to assess the association between environmental factors and the respective community, using the probability of occurrence of that community as the response variable. In all CRFs, the entire dataset was used to build the model rather than splitting it into training and test datasets, as this was consistent with the focus of the study, which was to infer the impact of environmental factors on benthic communities and not to predict communities' changes (Rykiel, 1996). We then assessed the relative importance of a given environmental factor in explaining the partitioning of communities or the probability of occurrence of a particular community by measuring the mean difference in accuracy (for general CRF: out-of-bag error; for individual CRFs: mean squared error) before and after conditional permutation for each tree. For each individual CRF, partial dependence plots were used to represent the marginal effects of each environmental factor on the probability of occurrence in a particular community. In this way, we were able to examine the nonlinear relationship between a given factor and the occurrence of a community (Cutler et al., 2007).

All means were presented as mean $\pm$ standard deviations. All analyses were performed using R v. 4.1.2 with the packages "adespatial" v. 0.3-16 (Dray et al., 2022), "ape" v. 5.6-2 (Paradis and Schliep, 2019), "bootcluster" v. 0.3.2 (Yu et al., 2018), "NbClust" v. 3.0.1 (Charrad et al., 2012), "olsrr" v. 0.5.3 (Hebbali, 2020), "party" v.1.3-11 (Hothorn et al., 2023), and "vegan" v. 2.6-2 (Oksanen et al., 2009).

Table 1

Justifications, ranges, and types of 21 environmental factors used for assessing their impacts on community delineation and occurrence.

Factor Type	Environmental Factor	Range	Justification
Physical	Mean sea surface temperature (Mean SST)	19.51–25.03 °C	Sea surface temperature has long been recognized as an important biophysical factor restricting the latitudinal distribution of tropical reefal species and coral reefs (Kleypas et al., 1999). Light intensity is an important biophysical factor restricting the vertical and latitudinal distribution of coral reefs (Tamir et al., 2019; Kleypas et al., 1999). Wave can cause the variation among benthic communities around reef islands (Lange et al., 2021). Chlorophyll <i>a</i> is a proxy for phytoplankton biomass and thus primary production (Fox et al., 2018), which generally influences the appearance and abundance of filter feeders (Cugier et al., 2010). Unstable substrate has a negative relationship with scleractinian coral cover and affects their choice of settlement choice and post-settlement survival (Sherman et al., 2016). Nutrient concentration affects the abundance of coral and algal species (De'ath and Fabricius, 2008). Extreme temperature anomalies and the subsequential recovery time from such events can change the abundance of zooxanthellate corals, thereby affecting the structure of benthic communities (Stuart-Smith et al., 2018). Recurrent and extreme typhoons, and the recovery time from typhoons can alter the abundance of benthic species (Ribas-Deulofeu et al., 2021)
	Standard deviation of sea surface temperature (SD SST)	1.69–3.28 °C	
	Light intensity	0.27–23.17 E × m ⁻² × year ⁻¹	
	Wave exposure	0.11–22.31 kW × m ⁻¹	
	Wave height	2.2–17 m	
	Mean chlorophyll <i>a</i> (Mean Chl <i>a</i>)	0.13–0.92 mg × m ⁻³	
	Standard deviation of chlorophyll <i>a</i> (SD Chl <i>a</i>)	0.02–0.48 mg × m ⁻³	
	Unstable substrate cover	0–52.5 %	
	Nitrate	0.003–0.709 μM	
	Nitrite	0.010–0.079 μM	
Chemical	Phosphate	0.013–0.007 μM	
	Degree heating week (DHW)	2.5–11.29 °C-weeks	
Historical	Recovery time from the last extreme heat event (DHW recovery)	2–13 years	
	Number of severe typhoons in the last decade (Typhoon frequency)	5–19 decade ⁻¹	
	Maximum wind speed for the last severe typhoon (Typhoon disturbance)	5.2–30.4 m × s ⁻¹	
	Recovery time from the last severe typhoon (Typhoon recovery)	0–4 years	
	Anthropogenic land use	0–265.36 km ²	
Land-based			Anthropogenic land was discovered to negatively relate to the

Table 1 (continued)

Factor Type	Environmental Factor	Range	Justification
	Forest land use	0–942.25 km ²	abundance of corals (Carlson et al., 2019). Forest land use generally positively influences the abundance of coral species (Carlson et al., 2019).
	Population density	0–5003 people × 10 km – radius ⁻¹	Human population density can affect benthic composition (Carlson et al., 2019).
	Tourist visits	0–10,396,664 visits	Increasing tourism stress can lead to a decline in the coral cover (Gil et al., 2015).
	Management status	Open/no-take/restricted	Regulations of marine protected areas can prevent the decline of coral cover (Magdaong et al., 2013).

3. Results

3.1. Community delineation and characteristics

Five communities were optimal for aggregating the 433 transects (best partitioning under the consensus of 26 indices; permutation test: $p < 0.01$; Appendix D). Assignment to a particular community was stable for 87 % of the transects (stability = 0.89 ± 0.09 ; Appendix E). Community occurrences were spatially more clustered than a random distribution ($p < 0.001$; Appendix F).

The Crustose Coralline Algae (CCA) community ($n = 100$) was dominated by CCA (44 ± 11 %), followed by turf algae (39 ± 9 %), and stony corals (9 ± 5 %; mainly with encrusting morphology), and generally thrived at all depths on the northeast coast of Taiwan and at 40 m depth in Kenting (Table 2; Fig. 1b). CCA, articulated calcareous algae, and encrusting and repent sponges were representative of this community (Table 1; Appendix G).

The Turf and Stony Coral community ($n = 128$) was dominated by turf algae (42 ± 11 %), encrusting stony corals (23 ± 7 %), and CCA (18 ± 7 %), and was found in relatively shallow waters (< 20 m depth) around Taiwan and offshore islands, with the exception of Xiaoliuqiu, where only two transects belonged to this community (Table 2; Fig. 1b). Column, encrusting, foliose, and massive stony corals as well as encrusting zoanthids were indicators of this community (Table 2; Appendix G).

Turf algae (66 ± 9 %), CCA (16 ± 6 %), and stony corals (12 ± 6 %, mostly with encrusting morphology) were abundant in the Turf community (Table 2), with >50 % of the 155 transects belonging to this community distributed around Xiaoliuqiu and Penghu, while others were scattered across different regions (Fig. 1b). Turfs, corticated foliose, and macrophyte algae characterized this community (Table 2; Appendix G).

Octocorals (38 ± 16 %), mainly digitate (29 ± 16 %) and lobate (6 ± 7 %), turf algae (26 ± 11 %), and CCA (21 ± 5 %) formed the Digitate Octocoral community ($n = 27$) (Table 2). This community was mainly distributed in the shallow waters of Dongsha, Kenting, and the east coast of Taiwan (Fig. 1b) and was typified by digitate, lobate, and massive octocorals, ascidians, and cyanobacteria (Table 2; Appendix G).

Finally, octocorals (32 ± 11 %) with bushy (20 ± 7 %) and clustered (11 ± 7 %) morphologies, turf algae (30 ± 11 %), and stony corals (23 ± 10 %) (Table 2) were ubiquitous in the Bushy Octocoral community ($n = 23$), which was found across depths in Ludao (Fig. 1b). It was characterized by arborescent, bushy, and table stony corals, arborescent,

Table 2

Morpho-functional composition of the five identified benthic communities.

Community	CCA	Turf & Stony Coral	Turf	Digitate Octocoral	Bushy Octocoral
Algae	3.2 ± 4.8	0.6 ± 0.8	2.1 ± 2.4	0.8 ± 0.8	0.3 ± 0.4
<i>Articulated calcareous</i>	2.1 ± 3.3*	0.2 ± 0.5	0.9 ± 1.1	0.4 ± 0.5	0.1 ± 0.1
<i>Corticated foliose</i>	0.8 ± 1.5	0.2 ± 0.5	0.9 ± 1.8*	0.3 ± 0.5	0.1 ± 0.2
<i>Corticated macrophyte</i>	0.4 ± 0.8	0.2 ± 0.2	0.4 ± 0.8*	0.1 ± 0.1	0.1 ± 0.3
Crustose coralline algae	43.5 ± 11.0*	18.4 ± 6.8	16.2 ± 5.5	21.0 ± 0.9	13.3 ± 6.1
Turf algae	39.3 ± 9.4	41.5 ± 11.4	65.8 ± 8.7*	26.4 ± 10.6	30.0 ± 10.6
Cyanobacteria	0.0 ± 0.1	0.1 ± 0.1	0.4 ± 1.4	0.1 ± 0.1*	0.0 ± 0.0
Stony corals	11.1 ± 6.5	35.2 ± 11.0	12.3 ± 6.2	13.1 ± 7.0	23.0 ± 9.8
<i>Arborescent</i>	0.0 ± 0.1	0.3 ± 0.9	0.1 ± 0.9	0.0 ± 0.1	2 ± 4.2*
<i>Bushy</i>	0.4 ± 0.8	3.3 ± 3.5	1.2 ± 1.8	1.0 ± 1.4	3.1 ± 3.1*
<i>Column</i>	0.1 ± 0.4	2.3 ± 3.7*	0.2 ± 0.5	0.1 ± 0.1	0.4 ± 0.5
<i>Encrusting</i>	9.4 ± 5.5	23.0 ± 7.0*	9.1 ± 4.7	10.1 ± 6.0	11.7 ± 4.7
<i>Foliose</i>	0.2 ± 0.4	1.0 ± 2.0*	0.2 ± 0.3	0.3 ± 0.4	0.4 ± 0.5
<i>Massive</i>	0.7 ± 1.1	3.0 ± 4.2*	1.0 ± 1.4	0.9 ± 0.9	1.9 ± 1.8
<i>Table</i>	0.2 ± 0.5	2.3 ± 4.6	0.5 ± 1.3	0.6 ± 0.8	3.4 ± 4.9*
Octocorals	0.7 ± 1.1	3.2 ± 3.3	1.8 ± 2.4	38.2 ± 16.4	31.8 ± 11.0
<i>Arborescent</i>	0.0 ± 0.0	0.1 ± 0.2	0.0 ± 0.0	0.0 ± 0.0	0.3 ± 0.4*
<i>Bushy</i>	0.2 ± 0.5	0.2 ± 0.5	0.3 ± 0.6	0.7 ± 0.8	20.0 ± 6.8*
<i>Clustered</i>	0.1 ± 0.3	0.3 ± 0.8	0.1 ± 0.3	1.0 ± 1.6	10.5 ± 6.6*
<i>Digitated</i>	0.3 ± 0.8	2.2 ± 2.8	1.1 ± 1.8	29.1 ± 16.3*	0.1 ± 0.1
<i>Encrusting</i>	0.0 ± 0.1	0.0 ± 0.0	0.0 ± 0.1	0.0 ± 0.0	0.8 ± 1.0*
<i>Lobate</i>	0.1 ± 0.2	0.3 ± 0.8	0.2 ± 0.6	5.9 ± 6.6*	0.0 ± 0.1
<i>Massive</i>	0.0 ± 0.1	0.1 ± 0.4	0.1 ± 0.2	1.4 ± 1.3*	0.1 ± 0.2
Sponges	1.7 ± 2.4	0.5 ± 0.6	1.1 ± 2.2	0.4 ± 0.4	1.6 ± 1.9
<i>Encrusting</i>	0.5 ± 0.9*	0.1 ± 0.3	0.4 ± 0.9	0.1 ± 0.1	0.4 ± 0.5*
<i>Massive</i>	0.3 ± 1.3	0.0 ± 0.1	0.3 ± 1.6	0.1 ± 0.1	1.1 ± 1.6
<i>Repent</i>	0.8 ± 1.2*	0.3 ± 0.6	0.5 ± 0.9	0.2 ± 0.3	0.1 ± 0.1
Zoanthids	0.3 ± 0.6	0.5 ± 1.6	0.3 ± 1.3	0.0 ± 0.1	0.0 ± 0.0
<i>Encrusting</i>	0.2 ± 0.6	0.4 ± 1.5*	0.2 ± 1.2	0.0 ± 0.1	0.0 ± 0.0
<i>Massive</i>	0.0 ± 0.0	0.0 ± 0.1	0.0 ± 0.1	0.0 ± 0.0	0.0 ± 0.0
Ascidian	0.0 ± 0.1	0.0 ± 0.0	0.0 ± 0.0	0.1 ± 0.1	0.0 ± 0.1
<i>Erect single</i>	0.0 ± 0.1	0.0 ± 0.0	0.0 ± 0.0	0.1 ± 0.1*	0.0 ± 0.1

Benthic percent cover (mean ± standard deviation) for each major benthic category (bold) and detailed cover for each morpho-functional group (italics). Asterisks denote the morpho-functional group indicators of each community identified by an indicator species analysis.

bushy, clustered, and encrusting octocorals, as well as encrusting sponges (Table 2; Appendix G).

The five communities were clearly differentiated from each other in the nMDS despite some overlap (Fig. 2; Appendix H). For the two algae-dominated communities, the CCA community had a higher proportion of CCA, articulated foliose algae, articulated calcareous algae, and repent sponges than the Turf community, which had a higher contribution of turf algae and massive sponges. As for the other coral-abundant communities, the Octocoral community was distinguished by a higher proportion of octocorals with digitated, massive and lobate morphologies. In addition, the Turf and Stony Coral community had a higher cover of encrusting and massive stony corals than the Bushy Octocoral community, which had a higher contribution of arborescent, bushy, column, and table stony corals, clustered, arborescent, and bushy octocorals (Fig. 2).

3.2. Environmental drivers of communities

The general CRF performed well overall (Appendix I), identifying light intensity, nitrite and SD Chl *a* as the most important factors for the community distribution. In contrast, mean SST, DHW recovery, and typhoon disturbance were identified as the least important factors influencing community distribution (Fig. 3a; Appendix J).

A moderate to high proportion of the variance was explained by individual CRFs (Appendix J), with distinct key environmental factors. Low light intensity, high typhoon frequency, high wave exposure, low DHW recovery, and management status (no-take zone) were associated with an increasing probability for the occurrence of the CCA community (Fig. 3b; Fig. 4a; Appendix J; Appendix K). The probability for the occurrence of the Turf and Stony Coral community was higher with low nitrite, high light intensity, high SD Chl *a*, low phosphate, and low anthropogenic land use (Fig. 3c; Fig. 4b; Appendix J; Appendix K). The probability of observing the Turf community was higher when unstable substrate, light intensity, and anthropogenic land use were high, tourist visitors were low, and management status was restricted (Fig. 3d; Fig. 4c; Appendix J; Appendix K). The probability of detecting the Digitate Octocoral community was greater when SD Chl *a*, population density, and nitrate were low, and when DHW and nitrite conditions were both high and low (Fig. 3e; Fig. 4d; Appendix J; Appendix K). The occurrence of the Bushy Octocoral community increased when the time to recover from typhoons is short, SD Chl *a*, and SD SST were low, and typhoon frequency and mean SST were high (Fig. 3f; Fig. 4e; Appendix J; Appendix K).

Thresholds that were characterized by a sigmoidal relationship between a number of environmental factors and community occurrences were: 1) light intensity with CCA, Turf and Stony Coral and Turf communities; 2) nitrite with Turf and Turf and Stony Coral communities; 3) typhoon frequency with CCA and Bushy Octocoral communities; 4) SD Chl *a* with Turf and Stony Coral and Digitate Octocoral communities; 5) wave exposure with Digitate and Bushy Octocoral communities (Fig. 3b; Appendix K).

4. Discussion

We find five communities with distinctive characteristics in the hard substrate environment around Taiwan, which are dominated by CCA, turfs, encrusting stony corals, and digitate and bushy octocorals. Several physical, chemical, and anthropogenic factors are important drivers of the difference in community composition and distribution, while the effects of historical factors are less noticeable for most communities. These findings demonstrate that the consideration of different baselines is particularly important for interpreting long-term dynamics and adapting management according to the local context to ultimately build climate-resilient communities (Gove et al., 2023).

The most influential physical factors for community delineation and their occurrence are light intensity, chlorophyll *a*, and wave exposure.

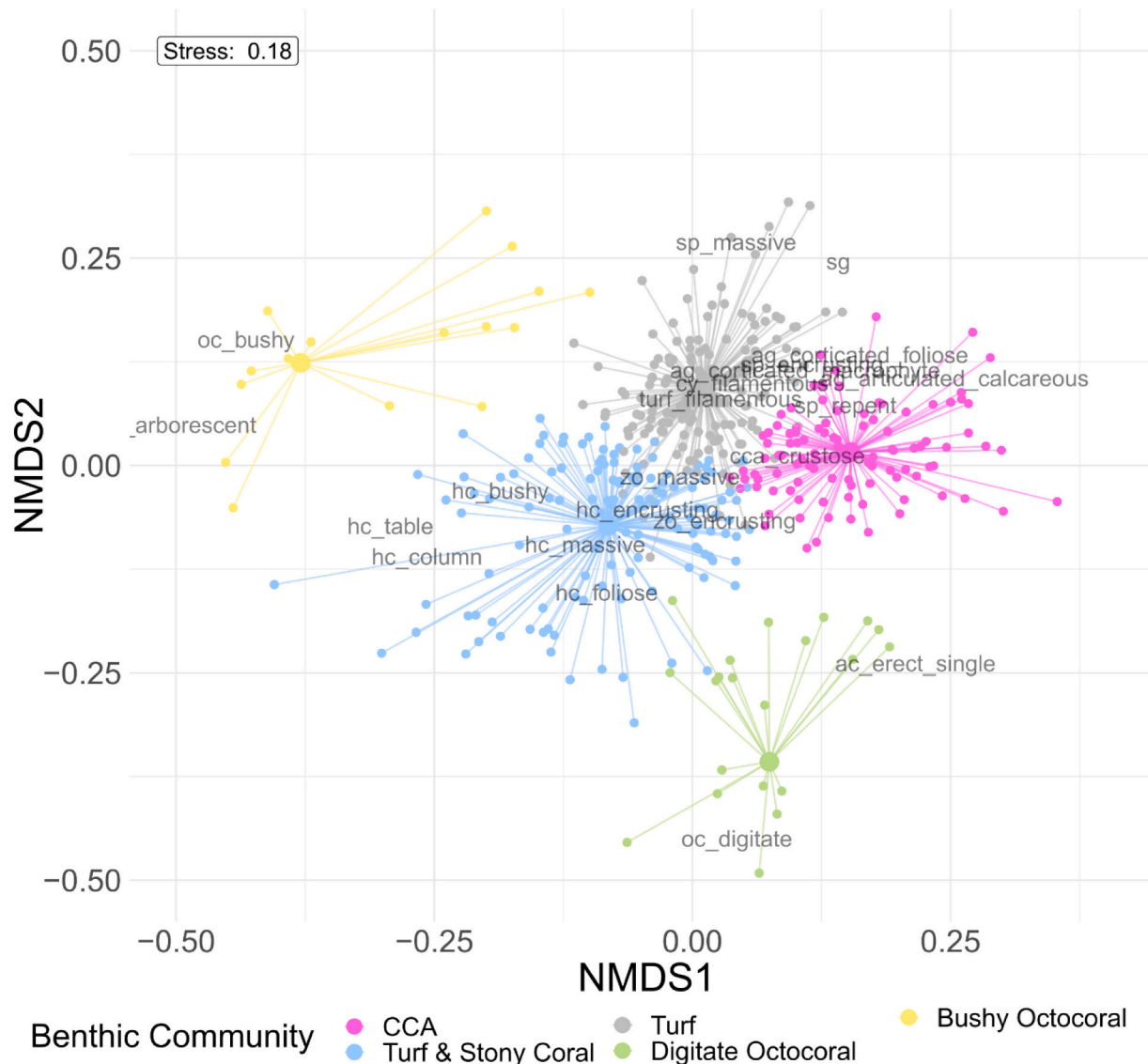


Fig. 2. nMDS showing similarity among transects with visualized benthic communities. nMDS based on the Euclidean distance of the Hellinger-transformed benthic matrix. The color denotes the communities identified by the *k*-mean cluster. The large size point indicates the centroid of each identified community, which is calculated as the centroid of all the transects (small points) belonging to the given community. The line segment denotes the ordination distance between the centroid of each community and the transects (small points) belonging to that community. The grey text represents the morpho-functional benthic categories. Abbreviation — ac: ascidians, ag: macroalgae, cca: crustose coralline algae, cy: cyanobacteria, hc: stony corals, oc: octocorals, sp.: sponges, sg: seagrass, zo: zoanthids. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

Light intensity critically influences the latitudinal and vertical distribution of autotrophic and mixotrophic organisms (Kleypas et al., 1999; Tamir et al., 2019), and apparently separates communities that thrive in high (Turf and Stony Coral, Turf, and Digitate Octocoral communities) and low light (CCA and Bushy Octocoral communities) environments with a clear threshold (between 5 and 15 $E \times m^{-2} \times yr^{-1}$). This confirms the earlier hypothesis about the key role of light in differentiating communities around Taiwan (Lin and Denis, 2019). While coral and algae-dominated communities generally prefer well-lit environments, previous studies suggest that xeniid and nephtheid corals, as well as CCA, favor lower light levels to avoid competition with other zooxanthellate benthic species (Shoham and Benayahu, 2016; Littler and Littler, 2013). The preference of the turf-dominated community to a well-lit environment indicates either the inherent benthic heterogeneity in tropical-subtropical reefs, or a sign of degradation introduced by other drivers (Szmant, 2012). Our findings suggest an association between the high chlorophyll *a* variability and Turf and Stony Coral

community, and a lower level is linked to the two octocoral-dominated communities. Chlorophyll *a* can represent food available to mixotrophic corals (Fox et al., 2018), therefore suggesting that pulses of heterotrophic nutrients could be important for stony coral-dominated reefs to increase their resilience to stressors (Fox et al., 2023). Alternatively, chlorophyll *a* also serves as an indicator of water quality in some regions since its positive correlation with turbidity and nutrient concentrations (De'ath and Fabricius, 2008). If this is true here, it could mean that the stony coral-dominated community is more resistant to poor water quality than the octocoral-dominated community. This could be possible, considering the main morphotypes that make up this community are encrusting and massive corals, many of which are known as stress-tolerant (Darling et al., 2012). It implies that the community has been or is developing under suboptimal conditions. Consequently, clarifying whether chlorophyll *a* indicates the food availability or water quality is critical for understanding the mechanism distinguishing stony- and octocoral-dominated communities. The observed effect of wave

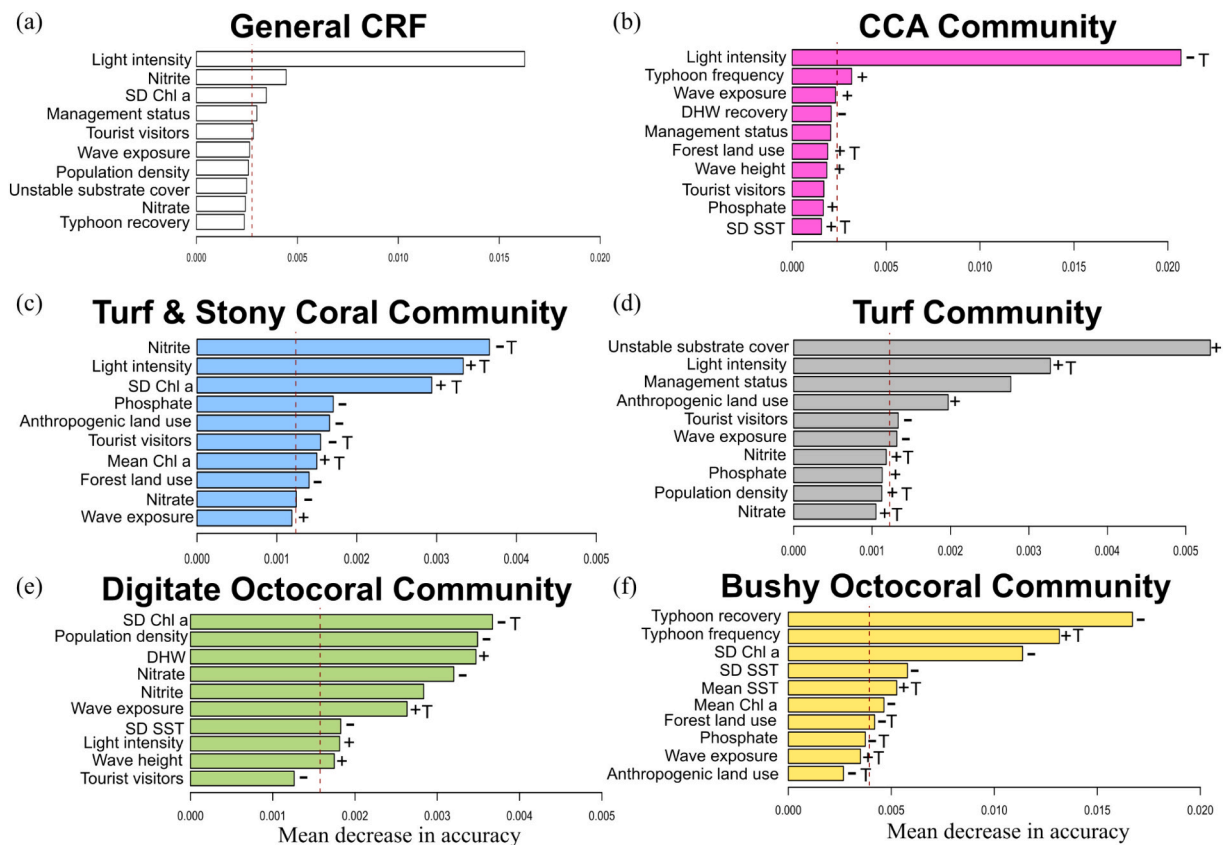


Fig. 3. Relative importance of the ten most important environmental factors for the distribution of communities as a whole and the occurrence of individual communities. (a) A general CRF. (b ~ f) CRFs of five individual communities. Positive and negative signs on the right side of the bar illustrate the general direction of the relationship between that environmental factor and the delineated community when it is distinguishable. The T sign on the right side of the bar indicates a sigmoid relationship between that environmental factor and the delineated community when it is distinguishable. Red vertical dashed lines indicate the mean importance of all environmental factors in the respective CRFs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

exposure on community distribution is reportedly consistent at different spatial scales (Porter et al., 2017; Hsiao et al., 2021; Lange et al., 2021). However, wave exposure surprisingly shows a negative association only with the Turf community, while it has a positive association with the other four communities. This contradicts previous findings across several Pacific Islands, where the turf-dominated community typically thrives in more exposed waters, reflecting the characteristic in wave tolerance of turf (Williams et al., 2013; Roff et al., 2018). Most of the physical factors used in the present study derived from remote sensing products, which are a clear advantage for working at a global/regional scale. However, they may not be suitable for capturing spatial and temporal variations in the local environment which may be important for some communities. Transitioning toward the use of *in situ* data when possible could be relevant for a better understanding of the variability observed at small spatial scale (e.g., Safaie et al., 2018; Hsiao et al., 2021).

Population density, tourist visits, and nitrite are parameters typically related to human activities and water quality, and are important in distinguishing between communities under fair water quality / low anthropogenic pressure (Turf and Stony Coral, Digitate and Bushy Octocoral communities) and communities with poor water quality / high anthropogenic pressure (Turf community). Elsewhere, these are widely recognized drivers of local shifts in benthic assemblages (Crisp et al., 2022), if propagated, lead to widespread reef degradation (Andrelo et al., 2021). A nitrite threshold (between 0.02 and 0.05 μM) differentiates these communities, and investigating whether this spatial threshold corresponds to a phase shift-inducing threshold (e.g., De'ath and Fabricius, 2008) would provide insight into the mechanism

underlying community dynamics. In contrast, nitrate is not identified here as a main factor for community delineation. Furthermore, no data on ammonium concentrations were available for this study. As different forms of nitrogen can have different sources and effects, it will be particularly important in the future to investigate the causes of nutrient enrichment and its effects on benthic organisms in more detail (Umezawa et al., 2008; Zhao et al., 2021).

In addition to the aforementioned factors, the Turf community is associated with an increase in unstable substrate, anthropogenic land use, phosphate, and nitrate. Although a turf-dominated community may occur naturally (Szmant, 2012), our results suggest that its prevalence is likely due to the long-term degradation of reefs in different regions of Taiwan. Degradation has been evidenced in some areas of Kenting and Penghu (southern and western Taiwan) over the past three decades, triggered by runoff from agriculture, tourist, and household wastewater (Dai, 1993; Hsieh et al., 2007; Meng et al., 2008; Huang et al., 2011; Liu et al., 2012). Similarly, Xiaoliuqi, an islet off the southwest coast of Taiwan, experienced a significant decline in coral assemblages concurrent with an increase in turfs (Yang et al., 1975; Dai, 2007), potentially induced by the coastal activities (Dai, 2007) and its proximity to highly eutrophic rivers (Liu et al., 2015). These regional degradations suggest an ongoing homogenization of communities toward turf-dominated across latitudes in Taiwan, with a few exceptions in the uninhabited areas (e.g., Dongsha Atoll). This is also in line with previous studies hypothesizing on the cause for the pervasiveness of turf in Taiwan (Dai, 2007; Ribas-Deulofeu et al., 2016) and is evidenced elsewhere, such as many Pacific islands (Jouffray et al., 2019; Smith et al., 2016). Moreover, the persistence of the turf-dominated community over past

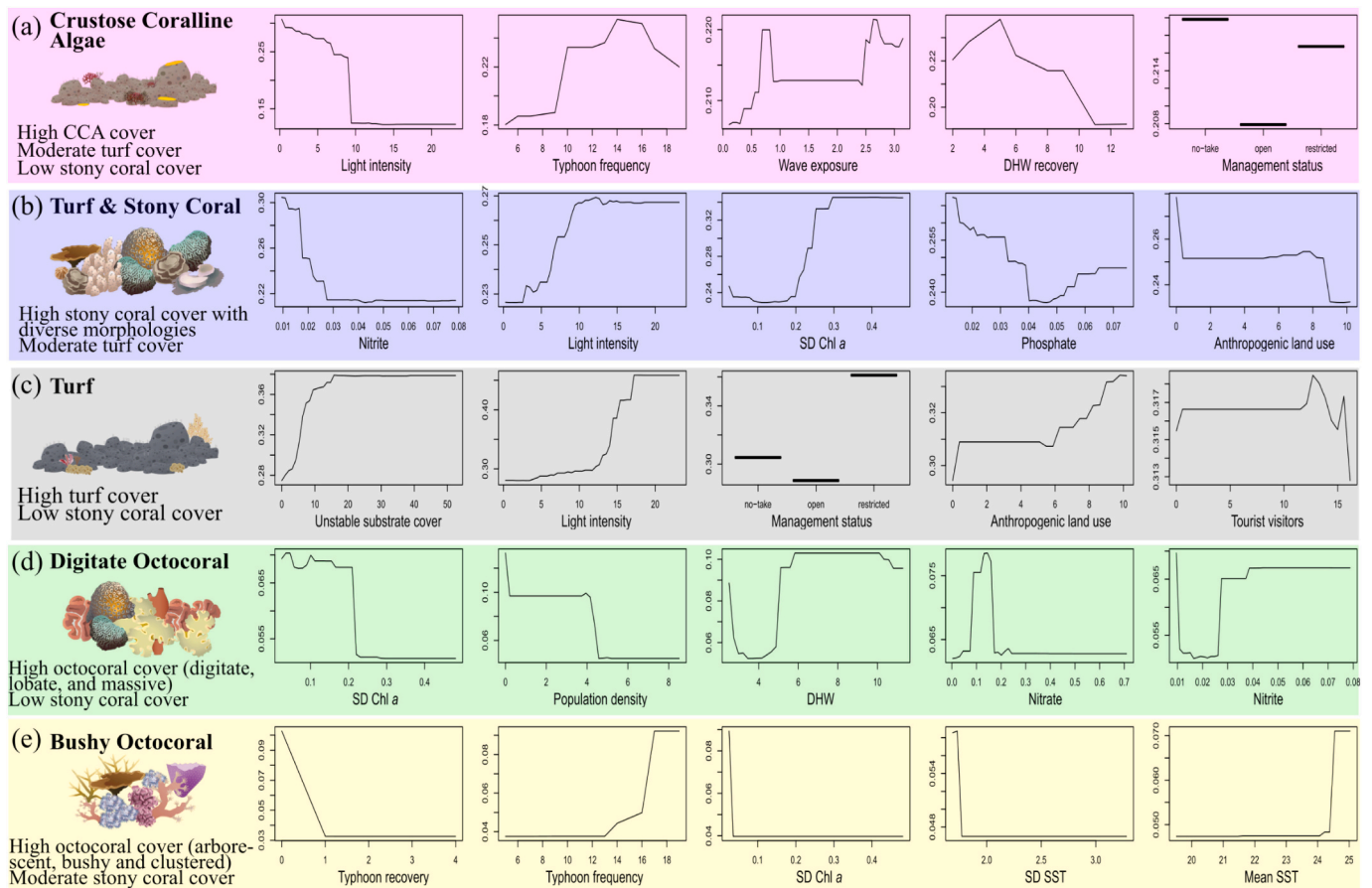


Fig. 4. Partial dependency of the top five environmental factors identified by individual CRFs and the dominant morpho-functional groups in each community. (a) CCA, (b) Turf and Stony Coral, (c) Turf, (d) Digitate Octocoral, (e) Bushy Octocoral communities. A schematic representation of the community is provided with its dominant groups (left-hand side), followed by the effects of environmental factors ordered from left to right according to their relative importance to the community occurrence. The units of environmental factors presented in the partial dependent plots are aligned with the corresponding units specified in Table 1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

decades may be explained by the low herbivory (Robinson et al., 2018), the chronic nutrient enrichment, and the high cover of the unstable substrate (current study). The latter could be due to coral mortality triggered by degradation, possibly triggering a harmful pathway that promotes the expansion of turf algae and hinders the settlement of coral larvae (Sherman et al., 2016). Eventually, management status emerges as the fourth most important factor in explaining the distribution of communities. Looking at the relationship with individual communities, it appears that the most stringent management efforts are placed in reefs with abundant turfs and CCA, rather than the ones with higher coral cover. Although the establishment purpose of some marine protected areas (MPAs) may target degraded reefs with high recovery potential (Abelson et al., 2016), it does not seem to be the case here, in Taiwan. Indeed, most Taiwanese MPAs have been implemented for years or decades, primarily targeting preserving biodiversity, habitats, or fisheries species. A majority of these MPAs (30 out of 46) adopt a fishery-oriented management strategy that regulates a limited number of economically important species (e.g., macroalgae, lobster, abalone). In other words, current protection aims to regulate human activities rather than exclude them. Accordingly, only a small proportion of MPAs are subject to 'no-take' status (18 % of the 87 locations surveyed, 80 % of which are in the remote and uninhabited Dongsha), which has been shown to provide the opportunity to effectively manage reefs (Magdaong et al., 2013). Future legislation should focus on ensuring ecosystem-level protection comprising no-take (core) zones and traditional use (buffer) zones, and the strict regulation on coastal human activities that can potentially lead to nutrient pollution and runoff. If rigidly implemented and enforced,

this would contribute to a comprehensive protection of reefs.

In contrast, most factors associated with typhoons and temperature anomalies play a minor role in shaping communities, with the exception of the Digitate and Bushy Octocoral communities. Because Taiwan is subject to frequent and severe typhoons (Mei et al., 2015), regular mechanical damage to reefs is evident in the morphology of the dominant corals (i.e., encrusting and massive; Ribas-Deulofeu et al., 2021). Therefore, the impact of typhoons may not be pronounced considering that these corals are widespread in the communities. On the other hand, the relatively low importance of thermal anomalies tends to go against the global trend, where the reefs with high coral abundance systematically match with the reefs with weak thermal disturbance (Darling et al., 2019). Their low ranks of importance suggest that the influence of certain chemical and land-based drivers may dominate, potentially distorting the role of environmental history and several biophysical factors (e.g., mean and SD SST) in community distribution as well as the regional differentiation. Furthermore, our study identifies typhoon impacts as crucial drivers for differentiating Digitate and Bushy Octocoral communities, with the former preferring sheltered reefs and the latter thriving in the waters exposed to typhoons. This challenges the hypothesis of Benayahu et al. (2004), suggesting an opposite pattern of preferred environmental conditions for the two communities, but is congruent with the finding in the Philippines (Lalas et al., 2023). Therefore, the increased typhoon frequency and intensity in the context of climate change (Mei et al., 2015) may further facilitate the persistence and expansion of the Bushy Octocoral community.

5. Conclusions

In summary, we identified five relatively homogeneous communities around Taiwan and revealed influential environmental drivers that affect their distribution. Their distinction is vital since they represent different baselines for interpreting their dynamic in the face of disturbances. We propose that human-induced effects may override the effects of natural drivers that typically shape these reef communities, converging many reefs in Taiwan into a community dominated by turf algae. Consequently, urgent action is needed to tailor management based on local baselines, implement strict regulations on coastal nutrients and runoff, and provide ecosystem-level protection. Such measures are essential to maintain community resilience against the worsening impact of climate change.

Credit authorship contribution statement

Yuting Vicky Lin: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Pierre-Alexandre Château:** Writing – review & editing, Methodology, Formal analysis. **Yoko Nozawa:** Writing – review & editing, Methodology, Formal analysis. **Chih-Lin Wei:** Writing – review & editing, Methodology. **Rainer Ferdinand Wunderlich:** Writing – review & editing, Methodology. **Vianney Denis:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Vianney Denis reports financial support was provided by National Science and Technology Council. Yuting Vicky Lin reports financial support was provided by Ocean Affairs Council of Taiwan. Yuting Vicky Lin reports financial support was provided by Marine National Park of Taiwan. Pierre-Alexandre Château reports financial support was provided by National Science and Technology Council. Chih-Lin Wei reports financial support was provided by National Science and Technology Council. Vianney Denis reports administrative support and travel were provided by Dongsha Atoll Research Station. Vianney Denis reports administrative support and travel were provided by Academia Sinica Green Island Marine Station. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data are publicly available from the Dryad Digital Repository <https://doi.org/10.5061/dryad.j0zpc86nz> (Lin et al., 2024). They are also made available together with the R script through the GitHub repository <https://github.com/NTU-FRELab/drivers-communities>.

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Appendices. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2024.116462>.

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